

The odd- c constant leaf of the SHARED-CONTENT LEMMA:

$$\text{content}(G_4^{(c)}) = 4 \text{ for odd } c \geq 7$$

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Abstract

Fix an odd integer $c \geq 7$ and let $G_4^{(c)}(a, b)$ be the polynomial in $\mathbb{Z}[a, b]$ obtained from the three-row engine $M_j = \langle s_{(a,b,c)}, h_1^{2(m-j)} e_2^j \rangle$ at $j = 4$ by peeling off the $2(c-5)$ “lower runs” (definition below, matching the even- c convention of [1]). We prove that

$$\text{content}(G_4^{(c)}) = 4,$$

where content is the minimum 2-adic valuation of the binomial-basis coefficients on the two odd-parity sheets $(\varepsilon_a, \varepsilon_b) \in \{(0, 1), (1, 0)\}$. The proof is a finite case check on the 50 coefficient polynomials of $G_4^{(c)}$ in $\mathbb{Z}[c]$: every coefficient has $v_2 \geq 4$ uniformly in odd c , and the coefficient $P_{1,1}^{(1,0)}(c) = 24c(25c^3 - 114c^2 + 131c - 12)$ has $v_2 = 4$ exactly, for every odd c .

1 Setup

Fix a three-row partition (a, b, c) with $a > b > c \geq 7$, c odd, and $a+b+c$ even (so that $|(a, b, c)| = 2m$ for an integer m). Since c is odd, $a+b$ is odd, so a and b have *opposite* parities.

The three-row engine polynomial at index j is

$$M_j = \langle s_{(a,b,c)}, h_1^{2(m-j)} e_2^j \rangle = \binom{2(m-j)}{b-j} \frac{(a-b+1) Q_c(a, b, j)}{c! (a+c+1-j) \prod_{i=1}^c (b+i-j)}. \quad (1)$$

The heavy-tip decomposition of Q_c is

$$Q_c(a, b, j) = (a - (c-2))(b - (c-1)) H_c(a, b, j) - (-1)^c (2c)! \binom{j}{2c}. \quad (2)$$

At $j = 4$ and $c \geq 3$, we have $4 < 2c$ so $\binom{j}{2c} = 0$, and $H_c(a, b, 4)$ is the polynomial part of $Q_c(a, b, 4)/((a-(c-2))(b-(c-1)))$. By the degree count of [1], $H_c(a, b, 4)$ has bi-degree $(c-1, c-1)$ in (a, b) . Dividing by the “lower runs” peels off $2(c-5)$ linear factors:

$$G_4^{(c)}(a, b) := \frac{H_c(a, b, 4)}{\prod_{s=3}^{c-3} (a+s) \cdot \prod_{s=2}^{c-4} (b+s)}. \quad (3)$$

Proposition 1 (Biquartic normal form; [1]). *For every integer $c \geq 6$, $G_4^{(c)}(a, b)$ is a polynomial in $\mathbb{Z}[a, b]$ of bi-degree $(4, 4)$. Its coefficients (as functions of c) are polynomials in $\mathbb{Z}[c]$ of degree at most 4.*

The proof of Proposition 1 in [1] does not use the parity of c : the bi-degree $(4, 4)$ statement follows from the degree count $(c - 1) - (c - 5) = 4$ (which holds for all $c \geq 6$), and integrality of the lower-runs quotient was verified there by symbolic interpolation for $c \in \{6, 8, \dots, 26\}$ and confirmed cross-computed against direct M_j evaluation. For the odd- c range relevant here ($c \in \{7, 9, \dots, 31\}$), integrality was verified explicitly during the present work.

1.1 The content on the parity sheet

For odd c , the parity sheets on which $s_{(a,b,c)}$ is a genuine Littlewood–Richardson coefficient (i.e., $a + b + c$ even) are indexed by $(\varepsilon_a, \varepsilon_b) \in \{(0, 1), (1, 0)\}$: substitute

$$a = 2u + \varepsilon_a, \quad b = 2v + \varepsilon_b, \quad (u, v) \in \mathbb{Z}_{\geq 0}^2.$$

Under this substitution, $G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)$ is a polynomial in (u, v) of bi-degree $(4, 4)$ with integer coefficients, admitting a unique binomial-basis expansion

$$P(u, v) = \sum_{i=0}^4 \sum_{k=0}^4 \alpha_{i,k} \binom{u}{i} \binom{v}{k}, \quad \alpha_{i,k} = (\Delta_u^i \Delta_v^k P)(0, 0).$$

The 2-adic content on the parity sheet is

$$\text{cont}_2(G_4^{(c)} \text{ on sheet } (\varepsilon_a, \varepsilon_b)) := \min_{i,k} v_2(\alpha_{i,k}) = \min_{(u,v) \in \mathbb{Z}^2} v_2(G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)).$$

The second equality is the standard Mahler / integer-Newton statement. We define

$$\text{content}(G_4^{(c)}) := \min \left\{ \text{cont}_2(G_4^{(c)} \text{ on sheet } (\varepsilon_a, \varepsilon_b)) : (\varepsilon_a, \varepsilon_b) \in \{(0, 1), (1, 0)\} \right\}.$$

2 Theorem statement

Theorem 2 (Odd- c constant leaf). *For every odd integer $c \geq 7$,*

$$\text{content}(G_4^{(c)}) = 4.$$

The upper bound (Section 5) exhibits a single universal witness that has $v_2 = 4$ exactly for every odd c . The lower bound (Section 6) is a finite check that every one of the 50 coefficient polynomials has $v_2 \geq 4$ uniformly.

3 The coefficient polynomials

Let $c \geq 7$ be odd. Fix a parity sheet $(\varepsilon_a, \varepsilon_b) \in \{(0, 1), (1, 0)\}$ and set

$$P_{i,k}^{(\varepsilon_a, \varepsilon_b)}(c) := (\Delta_u^i \Delta_v^k)[G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)]_{(u,v)=(0,0)}.$$

By Proposition 1, each is a polynomial in $\mathbb{Z}[c]$ of degree at most 4, obtained by interpolation from the values at, e.g., $c \in \{7, 9, 11, 13, 15\}$ (and cross-verified at $c \in \{17, 19, \dots, 31\}$, 0 mismatches).

3.1 Sheet $(\varepsilon_a, \varepsilon_b) = (0, 1)$

$(0, 0) : 48c(c-1)$	$(2, 3) : 2304(2c^3 + 18c^2 + 32c + 25)$
$(0, 1) : 24c(c-1)(5c^2 - 37c + 64)$	$(2, 4) : 9216(2c^2 + 6c + 5)$
$(0, 2) : 24c(c-1)(25c^2 - 121c + 150)$	$(3, 0) : 96c(c-2)(c-1)(9c - 7)$
$(0, 3) : 96c(c-2)(c-1)(9c - 7)$	$(3, 1) : 384c(c^3 + 16c^2 + 2c + 11)$
$(0, 4) : 384c(c-2)(c-1)(c+1)$	$(3, 2) : 2304(2c^3 + 18c^2 + 32c + 25)$
$(1, 0) : 24c(c-1)(5c^2 - 37c + 64)$	$(3, 3) : 9216(3c^2 + 18c + 25)$
$(1, 1) : 24c(25c^3 - 106c^2 + 117c - 44)$	$(3, 4) : 36864(2c + 5)$
$(1, 2) : 96c(9c^3 + 16c^2 - 36c + 33)$	$(4, 0) : 384c(c-2)(c-1)(c+1)$
$(1, 3) : 384c(c^3 + 16c^2 + 2c + 11)$	$(4, 1) : 1536c(c+1)(2c + 1)$
$(1, 4) : 1536c(c+1)(2c + 1)$	$(4, 2) : 9216(2c^2 + 6c + 5)$
$(2, 0) : 24c(c-1)(25c^2 - 121c + 150)$	$(4, 3) : 36864(2c + 5)$
$(2, 1) : 96c(9c^3 + 16c^2 - 36c + 33)$	$(4, 4) : 147456$
$(2, 2) : 192(2c^4 + 44c^3 + 90c^2 + 100c + 75)$	

3.2 Sheet $(\varepsilon_a, \varepsilon_b) = (1, 0)$

$(0, 0) : 0$	$(2, 3) : 2304(2c^3 + 20c^2 + 44c + 39)$
$(0, 1) : 24c(c-1)(5c^2 - 36c + 58)$	$(2, 4) : 9216(2c^2 + 10c + 13)$
$(0, 2) : 24c(c-1)(25c^2 - 93c + 74)$	$(3, 0) : 288c(c-3)(c-2)(3c - 5)$
$(0, 3) : 288c(c-1)(3c^2 - c + 2)$	$(3, 1) : 384c(c^3 + 14c^2 - 31c + 19)$
$(0, 4) : 384c(c-1)(c+1)(c+2)$	$(3, 2) : 2304(2c^3 + 16c^2 + 12c + 7)$
$(1, 0) : 120c(c-4)(c-3)(c-2)$	$(3, 3) : 27648(c^2 + 6c + 7)$
$(1, 1) : 24c(25c^3 - 114c^2 + 131c - 12)$	$(3, 4) : 36864(2c + 7)$
$(1, 2) : 96(9c^4 + 22c^3 - 28c^2 + 21c + 6)$	$(4, 0) : 384c(c-3)(c-2)(c-1)$
$(1, 3) : 384(c^4 + 18c^3 + 29c^2 + 39c + 18)$	$(4, 1) : 1536c(c-1)(2c - 1)$
$(1, 4) : 1536(c+1)(c+2)(2c + 3)$	$(4, 2) : 9216(2c^2 + 2c + 1)$
$(2, 0) : 120c(c-3)(c-2)(5c - 13)$	$(4, 3) : 36864(2c + 3)$
$(2, 1) : 96c(9c^3 + 6c^2 - 68c + 73)$	$(4, 4) : 147456$
$(2, 2) : 192(2c^4 + 44c^3 + 78c^2 + 52c + 39)$	

4 Basic 2-adic facts for odd c

For c odd and any integer polynomial $P(c) \in \mathbb{Z}[c]$, the reduction $P(c) \bmod 2^k$ depends only on $c \bmod 2^k$ (since $P(c_1) - P(c_2)$ is divisible by $(c_1 - c_2)$). This lets us determine $v_2(P(c))$ for all odd c by evaluating P modulo a fixed power of 2, using representatives $c = 1, 3, 5, 7, \dots$ up to the required residue class.

We use the following elementary facts throughout (all for c odd, $c \geq 7$):

Lemma 3. *For every odd integer c :*

$$(i) \ v_2(c) = v_2(c - 2) = v_2(c - 4) = v_2(c + 2) = 0.$$

(ii) $v_2(c-1) \geq 1, v_2(c-3) \geq 1, v_2(c-5) \geq 1, v_2(c+1) \geq 1$.

(iii) If $P(c) \in \mathbb{Z}[c]$ has odd constant term and every other coefficient even, then $v_2(P(c)) = 0$ for all c .

(iv) $c(c-1) \equiv 0 \pmod{2}$ and, more sharply, $v_2(c(c-1)) \geq 1$.

Proof. (i)–(ii) are direct: c odd forces $c-2$ and $c-4$ odd (they differ from c by an even number) and $c-1, c-3, c-5, c+1$ even. (iii) is the standard parity observation. (iv) follows from (i)–(ii). $\square$

5 Upper bound

Lemma 4 (Universal witness). *For every odd integer $c \geq 7$,*

$$v_2(P_{1,1}^{(1,0)}(c)) = 4.$$

Consequently, $\text{content}(G_4^{(c)}) \leq 4$.

Proof. From the coefficient table in Section 3, $P_{1,1}^{(1,0)}(c) = 24c(25c^3 - 114c^2 + 131c - 12)$. Write

$$R(c) := 25c^3 - 114c^2 + 131c - 12.$$

Then $v_2(P_{1,1}^{(1,0)}(c)) = v_2(24) + v_2(c) + v_2(R(c)) = 3 + 0 + v_2(R(c))$ (using c odd), so it suffices to show $v_2(R(c)) = 1$ for every odd c .

Reduce R modulo 4. For c odd, $c^2 \equiv 1 \pmod{8}$ (write $c = 2t + 1$; $(2t + 1)^2 = 4t(t + 1) + 1$ and $t(t + 1)$ is even), hence $c^2 \equiv 1 \pmod{4}$ and $c^3 = c \cdot c^2 \equiv c \pmod{4}$. Compute each term mod 4:

$$\begin{aligned} 25c^3 &\equiv c \pmod{4}, & \text{since } 25 &\equiv 1 \pmod{4}, \\ 114c^2 &\equiv 114 \cdot 1 \equiv 2 \pmod{4}, \\ 131c &\equiv 3c \pmod{4}, & \text{since } 131 &\equiv 3 \pmod{4}, \\ 12 &\equiv 0 \pmod{4}. \end{aligned}$$

Therefore

$$R(c) \equiv c - 2 + 3c - 0 = 4c - 2 \equiv 2 \pmod{4},$$

so $v_2(R(c)) = 1$ for every odd c . Thus $v_2(P_{1,1}^{(1,0)}(c)) = 3 + 1 = 4$. $\square$

6 Lower bound

Proposition 5. *For every odd integer $c \geq 7$ and every $(\varepsilon_a, \varepsilon_b) \in \{(0, 1), (1, 0)\}$ and every $(i, k) \in \{0, 1, 2, 3, 4\}^2$,*

$$v_2(P_{i,k}^{(\varepsilon_a, \varepsilon_b)}(c)) \geq 4.$$

Consequently, $\text{content}(G_4^{(c)}) \geq 4$.

We prove this by inspection of each of the 50 coefficients. To avoid repetition, we group them by the mechanism that provides the lower bound.

6.1 High-prefactor coefficients

For each coefficient $P_{i,k}^{(\cdot)}(c) = 2^\alpha \cdot Q(c)$ with $Q(c) \in \mathbb{Z}[c]$, if $\alpha \geq 4$ then $v_2(P) \geq 4$ trivially. This handles the following on *both* sheets (reading prefactors from Section 3):

$\alpha \geq 5$: (0,3), (0,4), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,0), (3,1), (3,2), (3,3), (3,4), (4,0), (4,1), (4,2), (4,3), (4,4).

These are 19 index pairs on each of the 2 sheets, i.e., 38 coefficients. The remaining 12 coefficients (with prefactor $2^3 \cdot m$ or $2^4 \cdot 3$, or the identically-zero coefficient $P_{0,0}^{(1,0)}$) require a targeted argument, given below.

6.2 Coefficients with prefactor $24 = 2^3 \cdot 3$

There are 8 coefficients with prefactor 24; they come in two shapes.

Shape A: $24c(c-1) \cdot Q(c)$ where $Q(c)$ is a quadratic with parity-preserving structure. On sheet (0,1): (0,1), (1,0), (0,2), (2,0). On sheet (1,0): (0,1), (0,2).

For c odd, $v_2(24) + v_2(c) = 3 + 0 = 3$ and $v_2(c-1) \geq 1$ (Lemma 3(ii)), so it suffices to show $v_2(Q(c)) \geq 0$ (automatic since $Q(c) \in \mathbb{Z}$).

Sharper: for these coefficients, $v_2(Q(c))$ is easily determined:

- Sheet (0,1), (0,1) = (1,0): $Q = 5c^2 - 37c + 64$. For c odd: $5c^2$ odd, $37c$ odd, 64 even $\Rightarrow Q \equiv \text{odd} + \text{odd} + \text{even} \equiv \text{even} \pmod{2}$. So $v_2(Q) \geq 1$, giving total $v_2 \geq 5$.
- Sheet (0,1), (0,2) = (2,0): $Q = 25c^2 - 121c + 150$. For c odd: $\text{odd} + \text{odd} + \text{even} \equiv \text{even}$. $v_2(Q) \geq 1$; total ≥ 5 .
- Sheet (1,0), (0,1): $Q = 5c^2 - 36c + 58$. For c odd: $5c^2$ odd, $36c$ even, 58 even $\Rightarrow Q$ odd. $v_2(Q) = 0$; total $= 3 + v_2(c-1) \geq 3 + 1 = 4$.
- Sheet (1,0), (0,2): $Q = 25c^2 - 93c + 74$. For c odd: $\text{odd} + \text{odd} + \text{even} \equiv \text{even}$. $v_2(Q) \geq 1$; total ≥ 5 .

All 6 coefficients of shape A satisfy $v_2 \geq 4$.

Shape B: $24c \cdot R(c)$ where $R(c)$ is a cubic. On sheet (0,1): (1,1). On sheet (1,0): (1,1).

For c odd, $v_2(24) + v_2(c) = 3$, so we need $v_2(R(c)) \geq 1$.

- Sheet (0,1), (1,1): $R(c) = 25c^3 - 106c^2 + 117c - 44$. Compute mod 4 (using $c^2 \equiv 1$, $c^3 \equiv c \pmod{4}$ for c odd):

$$25c^3 - 106c^2 + 117c - 44 \equiv c - 2 + c - 0 = 2c - 2 \pmod{4}.$$

For c odd, $2c - 2 = 2(c - 1)$ with $c - 1$ even, so $2c - 2 \equiv 0 \pmod{4}$. Hence $v_2(R) \geq 2$; total $v_2 \geq 3 + 2 = 5$.

- Sheet (1,0), (1,1): $R(c) = 25c^3 - 114c^2 + 131c - 12$; handled in Lemma 4, where we showed $R \equiv 2 \pmod{4}$ hence $v_2(R) = 1$. Total $v_2 = 4$. (This is the tight witness.)

6.3 Coefficients with prefactor $48 = 2^4 \cdot 3$

Only one: $P_{0,0}^{(0,1)}(c) = 48c(c-1)$. Since $v_2(48) = 4$, total $v_2 \geq 4$. ✓

6.4 Coefficients with prefactor $96 = 2^5 \cdot 3$

Already handled by Section 6.1.

6.5 Coefficients with prefactor $120 = 2^3 \cdot 15$

There are two, both on sheet $(1, 0)$: $(1, 0)$ and $(2, 0)$.

Sheet $(1, 0)$, $(1, 0)$: $P = 120 c (c - 4)(c - 3)(c - 2)$. For c odd, $v_2(120) + v_2(c) + v_2(c - 4) + v_2(c - 3) + v_2(c - 2) = 3 + 0 + 0 + v_2(c - 3) + 0 \geq 3 + 1 = 4$. ✓

Sheet $(1, 0)$, $(2, 0)$: $P = 120 c (c - 3)(c - 2)(5c - 13)$. For c odd, $5c - 13 = \text{odd} - \text{odd} = \text{even}$, so $v_2(5c - 13) \geq 1$; also $v_2(c - 3) \geq 1$. Total $v_2 \geq 3 + 0 + 1 + 0 + 1 = 5$. ✓

6.6 Coefficients with prefactor $288 = 2^5 \cdot 9$

Handled by Section 6.1 (prefactor 2^5).

6.7 Verification summary

Every one of the 50 coefficient polynomials $P_{i,k}^{(\varepsilon_a, \varepsilon_b)}(c)$ has $v_2 \geq 4$ for all odd $c \geq 7$:

- 38 coefficients are handled by the high-prefactor argument (Section 6.1): prefactor divisible by 2^5 (19 on each sheet).
- 1 coefficient has prefactor $2^4 \cdot 3 = 48$ (Section 6.3).
- 2 coefficients have prefactor $120 = 2^3 \cdot 15$ (Section 6.5), gaining the missing factor of 2 from $c - 3$ (odd c) or from $5c - 13$ (odd c).
- 6 coefficients of Shape A (Section 6.2), prefactor 24, gain $v_2 \geq 1$ from $c - 1$ and (in five of six cases) additional $v_2 \geq 1$ from the inner polynomial being even at odd c .
- 2 coefficients of Shape B (Section 6.2), prefactor 24, require the mod-4 analysis: sheet- $(1, 0)$ $(1, 1)$ gains exactly $v_2 = 1$ (giving total 4), sheet- $(0, 1)$ $(1, 1)$ gains $v_2 \geq 2$ (giving total ≥ 5).
- 1 identically-zero coefficient $P_{0,0}^{(1,0)} = 0$ (contributes ∞ and does not lower the content).

Total: $38 + 1 + 2 + 6 + 2 + 1 = 50$ index pairs, accounting for all coefficients on both sheets.

This proves Proposition 5.

7 Proof of Theorem 2

Combining Lemma 4 (upper bound content ≤ 4) and Proposition 5 (lower bound content ≥ 4):

$$\text{content}(G_4^{(c)}) = 4 \quad \text{for every odd integer } c \geq 7. \quad \square$$

8 Computational verification

1. **Extended empirical anchor.** $\text{content}(G_4^{(c)}) = 4$ was verified numerically for all odd $c \in \{7, 9, 11, \dots, 99\}$ (47 values) by two independent engines: (i) Lyra’s Pieri + Jacobi–Trudi route (uid 469, verified odd $c \in \{7, \dots, 31\}$ with double-anchor $G_4^{(7)}(2, 1) = 51408 = 2^4 \cdot 3213$), and (ii) the direct M_4 evaluator `compute/2026-07-21-odd-c-content-extension/direct_M4.py` (odd $c \in \{33, \dots, 99\}$), cross-checked 0/157 mismatch on three-row shapes with $|\lambda| \in [10, 22]$.
2. **Coefficient polynomials.** The 50 polynomials in Section 3 were interpolated from the 5 values $c \in \{7, 9, 11, 13, 15\}$ and cross-verified at $c \in \{17, 19, 21, 23, 25, 27, 29, 31\}$: 0/400 mismatch across the $50 \times 8 = 400$ interpolant-vs-actual comparisons.
3. **Coefficient-level v_2 .** Every coefficient’s minimum v_2 over odd $c \in \{7, 9, \dots, 99\}$ agrees with the case-analysis bound in Section 6. The tight witness $P_{1,1}^{(1,0)}(c)$ evaluates to $v_2 = 4$ at every odd c in the range, as predicted by the Lemma 4 mod-4 computation.

9 Consequences

Corollary 6 (Physical $\Delta(4)$ for odd c). *Using the physical identity $\Delta(4) = 2v_2(G_4^{(c)}) - 10$ (memory node 2026-07-05-general-c-gen4-cru-x-proved), we get*

$$\min \Delta(4) = 2 \cdot 4 - 10 = -2 \quad \text{for every odd } c \geq 7,$$

i.e., the generator $j = 4$ contributes a negative shift of 2 units below the tip-run scale on odd sheets. Contrast with the even- c leaf ([1]): $\min \Delta(4) \in \{0, 2\}$ depending on $c \bmod 4$.

Remark 7 (Sheet asymmetry and the -1 offset). *The odd sheets are asymmetric: $(1, 0)$ has content 4 while $(0, 1)$ has content 5. The K -witness of the even- c leaf, $K(c) = 24c(c-1)(c-4)(c-5)$, has $v_2 \geq 3 + 0 + 1 + 0 + 1 = 5$ for odd c ; naively this would predict content = 5. The observed content 4 therefore signals a genuinely new witness on sheet $(1, 0)$, and Lemma 4 identifies it: the cubic factor $25c^3 - 114c^2 + 131c - 12$ inside the $(1, 1)$ coefficient. Concretely, this cubic reduces to 2 (mod 4) for every odd c (a single-bit residue). The “ -1 offset” observed in the numerics is precisely the single-bit cancellation $v_2(K)_{\text{even}} - v_2(P_{1,1}^{(1,0)}) \geq 5 - 4 = 1$: the tight sheet $(1, 0)$ carries a coefficient that beats the K -witness by exactly one bit, uniformly in odd c .*

Remark 8 (Toward the SHARED-CONTENT LEMMA). *Theorem 2 together with [1] completes Instance 3 of the SHARED-CONTENT LEMMA in the two-tower form:*

$$\text{content}(G_4^{(c)}) = \begin{cases} \min(v_2(K(c)), 6) & \text{if } c \text{ even, } c \geq 6, \\ 4 & \text{if } c \text{ odd, } c \geq 7. \end{cases}$$

Instance 1 (Clio ♣ resonance, Lyra’s lane) and Instance 2 (uncapped two-tower, Lyra’s $\gamma(c)$) remain open at the time of writing.

Acknowledgments

Lyra independently rebuilt $G_4^{(c)}$ via Pieri and reported $\text{content}(G_4^{(c)}) = 4$ constantly on odd $c \in \{7, \dots, 31\}$ (uid 469, 2026-07-21), double-anchored at $c = 7$ via $51408 = 2^4 \cdot 3213$; that observation triggered the present PROVE cycle. The background probe extending the range to odd

$c \in \{33, \dots, 99\}$ (147 partitions cross-checked, 0 mismatch) sharpened the target from a suspected “ $c = 7$ hole” into a clean constant leaf. Rick’s abacus-blueprint suggestion (uid 454) previously motivated the finite-substitution style of proof used here.

References

- [1] Clio, *Lyra’s cap identity for $G_4^{(c)}$: $\text{content}(G_4^{(c)}) = \min(v_2(K(c)), 6)$ for even $c \geq 6$* , proof file `proofs/2026-07-20-lyra-cap-identity.tex`, 2026.
- [2] Clio, *General- c master valuation for the three-row heavy engine*, memory node `2026-06-17-generalc-master-and-c5`, 2026.